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LA Community Health Expansion Index

Identifying high-priority ZIP codes for new community health centers across Los Angeles County

640

ZIP Codes Analyzed

365

ZIPs With No Nearby Clinics

12.4%

Avg Poverty Rate

1.000

Max Priority Score



What is a Federally Qualified Health Center?

**Community
Based**

**Provide Comprehensive
Primary Care**

Non-Profit

**Accept
Medi-Cal & Medicare**

**Offer Sliding-Scale Fees
for Uninsured Patients**

Safety Net for Underserved Communities

Why Does Targeted Health Expansion Matter in LA County?

Healthcare Deserts

365 of 640 ZIP codes analyzed (57%) have zero nearby community health clinics, leaving residents with no accessible primary care.

Concentrated Poverty

Top-priority ZIPs average over 40% poverty rates — far above the county average — yet receive the least healthcare infrastructure investment.

No Unified Prioritization

Without a data-driven index, expansion decisions rely on intuition. Resources go to visible areas, not necessarily the neediest ones.

How the index was built

A transparent, reproducible scoring methodology

01

Data Collection

Merged ACS census data & CDC PLACES health outcome data, and FQHC clinic location data for all LA County ZIP codes.

02

Feature Engineering

Created community-level indicators related to socioeconomic conditions, health outcomes, and healthcare accessibility.

03

Model Development(Built two expansion index models)

1. v1 used ACS data with a Weighted Composite Index and manually assigned weights (Healthcare Need 45%, Demand 30%, Socioeconomic Factors 25%).
2. v2 used ACS data and CDC PLACES health outcome data with Ridge Regression-derived weights to prioritize ZIP codes.

04

Priority Ranking

Generated final ZIP code rankings based on estimated healthcare need and clinic accessibility. ZIP codes with higher need and lower healthcare access ranked highest.

ACS Indicators of Community Need

Socioeconomic and demographic indicators used to measure healthcare need

NEED / VULNERABILITY

Variables associated with economic hardship and elevated healthcare need



poverty_rate

% below federal poverty line



unemployment_rate

Labor market distress proxy



median_household_income

Inverse income-based vulnerability



pop_hispanic + pop_black

Historically underserved populations



age_dependency_ratio

Elderly + child burden on working adults

DEMAND / POPULATION

Population-based indicators of healthcare service demand



population

Total resident population



percent_household_with_kids

Family-centered healthcare demand



household_size

Household crowding / utilization proxy



median_age

Age-related healthcare demand

SOCIOECONOMIC CONTEXT

Environmental and access-related barriers impacting care utilization



edu_highschool_percent

Health literacy proxy



median_age

Senior care demand indicator



mean_commute_time

Transportation access barrier



These ACS indicators help capture who is most vulnerable, where demand for healthcare is highest, and the structural barriers that affect access to care across LA County ZIP codes.

PLACES Health Burden Indicators

PLACES variables capture community health burden and preventive care utilization across LA County ZIP codes

CHRONIC DISEASE & HEALTH BURDEN

Higher values indicate elevated community healthcare need



ACCESS2

Uninsured population rate



DIABETES

Diagnosed diabetes prevalence



BPHIGH

High blood pressure prevalence



CHD + STROKE

Cardiovascular disease burden



COPD + CASTHMA

Respiratory disease burden



KIDNEY

Chronic kidney disease



OBESITY + LPA

Lifestyle-related health risk



MHLTH + PHLTH

Poor mental & physical health days



CSMOKING

Current smoking prevalence



CASTHMA

Current asthma



TEETHLOST

All teeth lost (oral health neglect proxy)

PREVENTIVE CARE UTILIZATION

Lower utilization may indicate healthcare access gaps



CHECKUP

Annual primary care utilization



DENTAL

Preventive dental care access



CERVICAL

Cervical cancer screening



COLON_SCREEN

Colorectal cancer screening participation



MAMMOUSE

Mammography utilization



PLACES variables helped capture both underlying disease burden and access to preventive healthcare services across LA County ZIP codes.

Top 10 Priority ZIP Codes

v1: ACS socioeconomic data using a Weighted Composite Index

#	ZIP	Score	Pop.	Poverty
1	92258	1.000	627	53.1%
2	90003	0.976	75,190	26.5%
3	92283	0.934	1,918	46.8%
4	90011	0.933	109,511	27.3%
5	92281	0.928	2,548	37.2%
6	90059	0.923	41,315	22.7%
7	90002	0.901	54,384	26.9%
8	91963	0.900	501	25.0%
9	90044	0.900	100,225	27.6%
10	92327	0.890	756	66.8%

v2: ACS + CDC PLACES health outcome data using Ridge Regression–derived weights

#	ZIP	Score	Pop.	v1 rank	change
1	92327	1.000	756	#10	↑9
2	92258	0.970	627	#1	↓1
3	92283	0.796	1,918	#3	—
4	92281	0.791	2,548	#5	↑1
5	90003	0.787	75,190	#2	↓3
6	90073	0.784	881	#280	↑274
7	92274	0.780	12,843	#54	↑47
8	92347	0.772	754	#12	↑4
9	90011	0.768	109,511	#4	↓5
10	90058	0.766	3,659	#13	↑3

Key Insights

What the results tell us about potential FQHC expansion

01

ZIP 90073: High Need, Low Access

Moved from #280 to #6 after adding PLACES data.
0 nearby clinics + high diabetes, obesity, and inactivity rates suggest major unmet healthcare need.

02

ZIP 92274: Hidden Expansion Need

Moved from #54 to #7 in v2.
Moderate poverty rate, but large population, no nearby clinics, and elevated health burden.

03

6 / 10 ZIPs Stayed in the Top 10

Both models identified similar high-need areas.
PLACES data improved prioritization precision.

04

Poverty Alone Is Not Enough

Poverty rate strongly correlates with the expansion index ($r = 0.761$), but healthcare need also depends on health outcomes and clinic access.

05

Overall Insight

The final model helps identify ZIP codes with both high healthcare need and limited clinic accessibility for future FQHC expansion.

Interactive Dashboard

A Streamlit app that turns the index into an explorable decision tool:

<https://lahealthexpansionindex.streamlit.app>



Recommendations

Actionable next steps for health system planners and policymakers

01

Extreme Need Priority Areas: 92327, 92258

These ZIP codes show extremely high healthcare vulnerability, severe poverty levels, and limited clinic accessibility despite smaller populations.

02

High Population Expansion Opportunities: 92274, 90003

These ZIP codes combine elevated healthcare need with larger populations, allowing future FQHC expansion to potentially serve more residents.

03

Use the Index for Grant Applications

The expansion score provides quantitative justification for HRSA and state FQHC designation applications — a competitive edge in funding.

04

Integrate Real-Time Clinic Data

The current index uses a static clinic snapshot. Connecting to live HCAI or 211 LA data would keep the rankings current as infrastructure changes.

05

Embed the Dashboard in Planning Workflows

Share the Streamlit app with county health officials and community partners so decisions are data-driven by default, not just during reviews.



Thank
you very
much!

